Project Title
Local Business Intelligence & Analytics Platform

ELECTRONICS SALES

1. Project Overview

This report presents an exploratory data analysis (EDA) of an electronics sale's dataset. It allows users to load data, perform analysis, visualize results, and generate reports. The objective of this analysis is to understand sales patterns, identify top-performing products, and extract meaningful business insights to support data-driven decision-making.

2. Project Architecture

Modules:

- data_loader: Load & clean data
- core_logic: Menu + revenue
- analysis: insights
- visualization: charts

3. Dataset Overview

It Contains sales records such as product, amount, and date. The dataset contains transaction-level sales data including date, product, customer, and revenue fields. Initial inspection included previewing rows, checking column names, and identifying missing values.

File: electronics_sales.csv
Total Rows: 4203
Total Columns: 6
Columns: order_id, date, customer, product, category, amount

4. Data Cleaning

Data cleaning steps included:

- Converting the date column into datetime format
- Handling missing or inconsistent values
- Ensuring numeric columns were properly typed

5. Exploratory Data Analysis

5.1 Revenue Analysis

Total revenue, average order value and total transactions were calculated to understand business performance. These metrics provide a quick summary of overall sales.

Revenue Analysis
Total Revenue: 209307999.0
Average Order Value: 49799.666666666664
Total Transactions: 4203

5.2 Product Analysis

A frequency analysis was conducted to identify the most frequently sold products. This helps the business understand which products drive most transactions.

Product-wise analysis was conducted to identify:

Top selling products

product	
Smartwatch	647
Speaker	625
Laptop	616
Headphones	598
Tablet	578

Least selling products

product	
Camera	562
Mobile	577

Tablet	578
Headphones	598
Laptop	616

Category-wise Sales

category	
Electronics	199514976.0
electronics	9793023.0

5.3 Customer Analysis

Customer-level analysis helped identify unique customers, most active customers , high-value customers, purchase frequency, and overall contribution to revenue.

Total Unique Customers: 795

Most Active Customers

customer	
Customer_299	15
Customer_514	14
Customer_631	13
Customer_183	13
Customer_372	13

5.4 Sales Trend Analysis

Sales data was grouped by date to observe trends over time. This analysis helps in identifying daily sales trend, monthly sales trend, growth trends or unexpected fluctuations in revenue.

Daily Sales Trend

date	
2026-01-01	2307079.0
2026-01-02	1441983.0
2026-01-03	1204476.0
2026-01-04	1402211.0
2026-01-05	1764009.0

Monthly Sales Trend

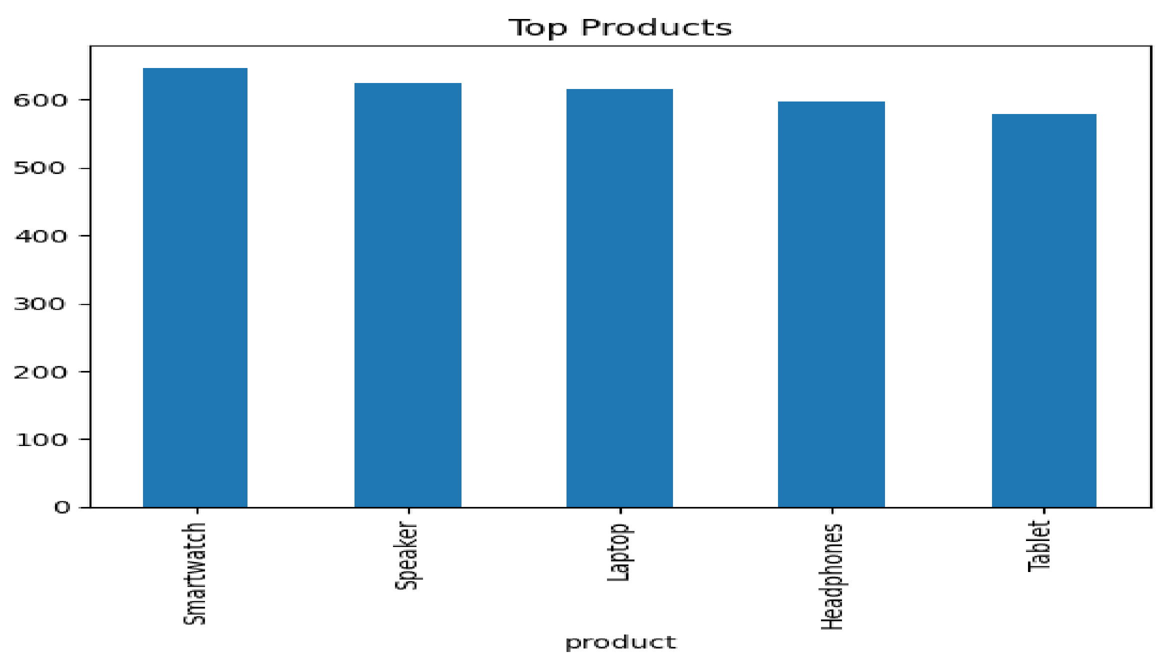
date	
2026-01	56043788.0
2026-02	45630173.0
2026-03	54606743.0
2026-04	51488034.0
2026-05	1539261.0

6. Visual Insights

Bar charts were used to visualize the top-selling products, while line charts were used to illustrate sales trends over time and the pie chart illustrates how sales amount is distributed across each category. Visualizations make it easier to communicate patterns and insights to stakeholders.

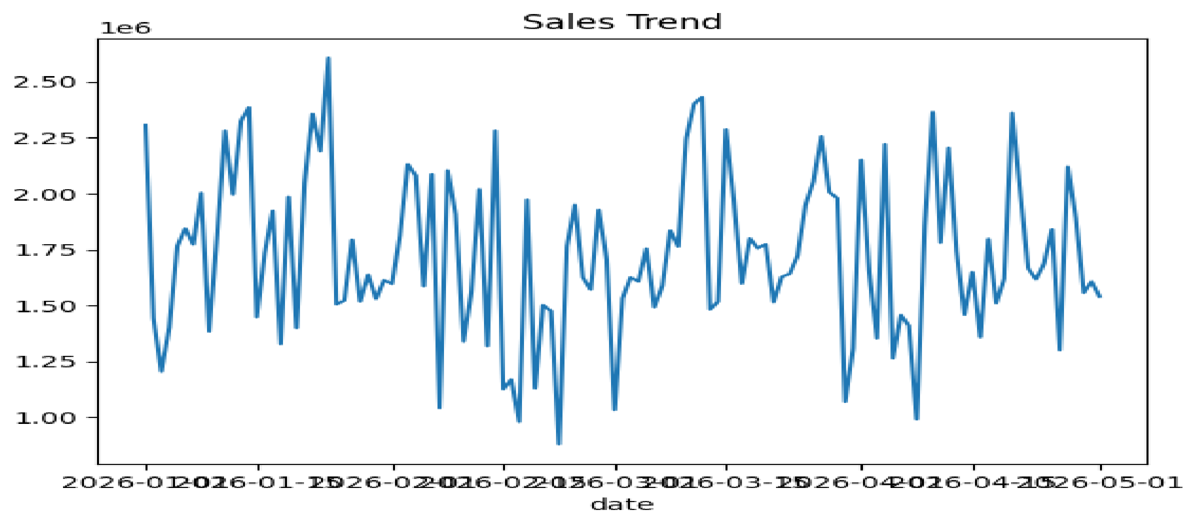
Top Selling Products

The following chart shows the products with the highest number of transactions, helping identify items that contribute most to sales volume.



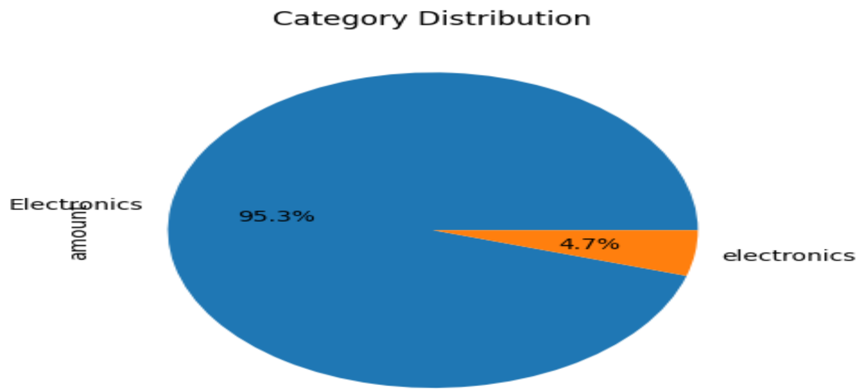
Sales Trend Over Time

This line chart illustrates how sales revenue changes over time. It helps identify growth patterns, seasonal effects, and any unusual fluctuations in performance.



Category wise sales

The pie chart illustrates how sales revenue is distributed across each category. It helps identify the percentage share of each category.



7. Conclusion

The exploratory data analysis provided a clear understanding of product performance and sales trends. These insights can help the organization optimize inventory, improve marketing efforts, and make informed business decisions based on data.